Peng Li

Acoustic Modelling | Computational Fluid Dynamics | Fluid–Structure Interaction

Stockholm, Sweden | +46 79 010 31 96 | pengli.2810@gmail.com

ORCID: 0009-0006-0575-0865

Professional Summary

Ph.D. researcher at KTH Royal Institute of Technology specialising in computational acoustics, wave propagation, and coupled fluid–structure interaction (FSI). Develop analytical acoustic models and high-fidelity CFD–FEM simulations to explain sound generation, duct resonances, and flow-induced vibration. Experience includes compressible large eddy simulation (LES), acoustic-source decomposition, spectral analysis, scientific programming, and simulations on Swedish HPC infrastructure. Bring a foundation in applied mathematics, fluid mechanics, and structural dynamics relevant to acoustic modelling and signature analysis in underwater applications. **Ph.D. defence scheduled for 4 December 2026.**

Technical Skills

Acoustic modelling: Convected wave equations; duct acoustics; acoustic resonances; mode matching; evanescent modes; junction impedance; acoustic–structural coupling; Ffowcs Williams–Hawkings (FW–H) analogy; duct Green's functions.

CFD and FSI: Compressible LES; finite-volume and finite-element methods; strong two-way coupling; overset and deforming meshes; hyperelasticity; flow-induced vibration; transient compressible flow and shock-wave propagation.

Signal and modal analysis: FFT; power spectral density; cross-spectra; coherence and phase; spectral proper orthogonal decomposition (SPOD); wavenumber–frequency analysis; structural eigenmodes.

Scientific computing: Python, MATLAB, C/C++, Fortran, Mathematica; parameter studies; model verification and validation; mesh and time-step convergence; HPC simulation workflows (NAISS resources at PDC and NSC).

Engineering software: Simcenter STAR-CCM+, OpenFOAM, COMSOL Multiphysics, ABAQUS, ANSYS, Tecplot, Gnuplot, Inkscape, LaTeX.

Research and Professional Experience

Ph.D. Researcher – Computational Aeroacoustics and FSI Mar 2022 – Present

Department of Engineering Mechanics, FLOW, KTH Royal Institute of Technology, Stockholm

- Develop three-dimensional upper-airway models coupling compressible LES with FEM-based soft-tissue mechanics in STAR-CCM+, using strong two-way FSI, Neo-Hookean materials, and moving overset meshes.
- Develop an analytical wave-propagation and resonance model for a bifurcated duct with a compliant splitter, combining convected plane waves, mode-matched junction impedance, and frequency-dependent structural compliance.
- Verify acoustic-model predictions against COMSOL eigenfrequency calculations: frequency differences within 0.002% for 27 rigid-splitter resonances and 1.03% for the first ten soft-splitter common resonances in the tested geometry (unpublished manuscript).
- Evaluate monopole, dipole, and quadrupole sound-source contributions using the FW–H analogy and duct Green's functions; relate acoustic pressure spectra to unsteady loading, structural motion, and breathing conditions.
- Use pressure/vorticity SPOD, wavenumber–frequency analysis, cross-spectra, coherence, and modal analysis to distinguish acoustic standing waves from convective flow structures and identify fluid–structure coupling mechanisms.
- Conduct mesh and time-step convergence studies and simulations using NAISS HPC resources at PDC and NSC; lead analysis, visualisation, and manuscript writing in collaboration with researchers in fluid mechanics, FSI, and acoustics.

Research Assistant – Transient Modelling and Compressible CFD Dec 2018 – Mar 2020

Joyson Safety Systems, Aschaffenburg, Germany

- Developed Python models of pyrotechnical initiators from mass and energy conservation, combustion, heat transfer, and nozzle-flow relations.
- Coupled 0-D model outputs to 3-D OpenFOAM simulations to resolve transient shock-wave propagation and assess airbag-inflator performance and structural integrity.
- Compared predicted pressure histories with closed-bomb measurements and performed parameter-sensitivity studies to support product development; work published in *Journal of Energy and Power Technology*.

Education

Ph.D. studies – Computational Fluid Dynamics Mar 2022 – Present
KTH Royal Institute of Technology, Stockholm, Sweden
Thesis: *Computational Aero-acoustics of Snoring with Impact on Obstructive Sleep Apnea Diagnosis*
Defence scheduled: 4 December 2026.
Supervision: Mihai Mihaescu, Marco Laudato, and Susann Boij.

M.Sc. in Mechanics (Mechanik) Oct 2016 – Jan 2020
Technical University of Darmstadt, Germany
Final grade: **1.33 (very good)**; thesis: **1.0** (German grading system).
Thesis: *Analysis and modeling of ballistic behaviour of pyrotechnical initiators and their impact in airbag inflators applications.*

B.Eng. in Engineering Mechanics Sep 2011 – Jun 2015
Dalian University of Technology, China
Overall mark: 86/100; thesis mark: 98/100.

Selected Peer-Reviewed Journal Publications

1. **Li, P.**, Laudato, M., and Mihaescu, M. (2026). Computational analysis of palatal non-apneic snoring sound generation using a simplified human upper-airway model. *Physics of Fluids*, **38**(8), 081909. **Featured Article**. doi:10.1063/5.0346307.
2. **Li, P.**, Laudato, M., and Mihaescu, M. (2023). Time-dependent fluid-structure interaction simulations of a simplified human soft palate. *Bioengineering*, **10**(11), 1313. doi:10.3390/bioengineering10111313.
3. **Li, P.**, Chang, C.-Y., and Sadiki, A. (2021). Numerical investigation of pyrotechnical initiators and their impact in airbag inflator applications: Ballistic modeling and functional analysis. *Journal of Energy and Power Technology*, **3**(2), article 026. doi:10.21926/jept.2102026.
4. Wang, B., **Li, P.**, and Li, R. (2016). Symplectic superposition method for new analytic buckling solutions of rectangular thin plates. *International Journal of Mechanical Sciences*, **119**, 432–441. doi:10.1016/j.ijmecsci.2016.11.006.

Unpublished Manuscripts

1. **Li, P.**, Laudato, M., Boij, S., and Mihaescu, M. *Acoustic resonances of a bifurcated duct with a compliant splitter.* Analytical wave modelling, junction impedance, acoustic–structural coupling, and FEM verification.
2. **Li, P.**, Laudato, M., and Mihaescu, M. *Effects of tongue-base induced airway constriction on non-apneic palatal snoring.* Coupled CFD–FSI, pressure/vorticity SPOD, and separation of hydrodynamic and acoustic modes.
3. **Li, P.**, Laudato, M., and Mihaescu, M. *Stiffness-dependent modal excitation and force–velocity coupling in fluid-induced soft-palate vibration.* Modal coupling, nonlinear vibration, and sound-source analysis.

Additional Research Experience

Research Assistant – Reduced Combustion Modelling Sep 2020 – Aug 2021
Institute for Technical Thermodynamics, Karlsruhe Institute of Technology, Germany
Applied the reaction–diffusion manifold (REDIM) method to premixed flame–wall interactions; validated reduced kinetic models against detailed kinetics.

Research Project – Low-Mach-Number Flow Modelling Aug – Sep 2018
Institute for Fluid Dynamics, Technical University of Darmstadt, Germany
Used COMSOL FEM to investigate non-isothermal Couette flow and natural convection, comparing low-Mach-number and Boussinesq formulations.

Research Assistant – Analytical and Numerical Solid Mechanics Jul 2013 – Jun 2015
Institute for Solid Mechanics, Dalian University of Technology, China
Derived analytical plate bending and buckling solutions and compared them with FEM using Mathematica, ABAQUS, and MATLAB.

Scientific Communication

Co-authored *Effects of gravity force and tissue stiffness on a simplified three-dimensional human soft palate*, ICSV30, Amsterdam (2024). Conference participation: EFDC2, Dublin (2025); Swedish Mechanics Days, Gothenburg (2024); Swedish Society of Biomechanics, Lund (2024).

Languages

Chinese – Native | English – IELTS 6.5/9.0 | German – TestDaF 17/20
Swedish – A2 (CEFR)